GEK-83946A

Instructions

SILCOMATICB Plus 6 Pulse Firing

Static Kraemer Adjustable Speed Drive With DIRECTO-MATIC® Plus Hardware

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NOTE — Commands attention to an essential operating or maintenance procedure, condition, or statement that must be highlighted.

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CHAPTER 1

INTRODUCTION

1-1. GENERAL INFORMATION

The purpose of this instruction book is to provide sufficient information to install, maintain and trouble-shoot a SILCOMATIC® Plus 6-pulse Kraemerstat. These instructions are as general as possible, but they have been designed to supply information that is only required on a specific job. It is written for use by all personnel involved with the power converter (owners, operators, field engineers, electricians, technicians, other maintenance personnel, etc.). Some knowledge of electrical theory and experience in electrical technology is required to implement these instructions. This is not the only information necessary to install a complete system. Information and instructions on the process control equipment, interface cabling, etc. are also required, including the specific power converter elementaries. These instructions do not propose to cover every contingency of installation, maintenance, and trouble-shooting. Should further information be desired or particular problems arise which are not sufficiently covered here, they should be referred to General Electric Company. Recommended changes, additions or deletions to these instructions should be addressed to:

Drive Systems Department
Attn: Power Conversion Engineering
1501 Roanoke Blvd.
Salem, Virginia 24153
Telephone: DIAL COMM 8*278-8672
AC 703-387-8672

CHAPTER 2

SYSTEM DESCRIPTION - POWER CIRCUIT

2-1. FUNCTIONAL DESCRIPTION

The Static Kraemer Drive (SKD) is a complete rectifier system which converts ac rotor power to dc and inverts this to line frequency at line voltage for feedback into the ac system. By varying the firing angle of the inverter, more or less voltage will oppose the dc voltage. As inverter voltage is lowered, higher dc current will flow with consequently higher rotor current. This will increase the torque produced and increase the speed of the rotor. Effectively then, changing the firing angle changes the rotor resistance which changes the speed.

The regulating system consists of an inner current loop and an outer speed loop. These two together provide a smooth adjustment to control the speed over the operating range (50 to 98.5% of synchronous speed of 1000 RPM). Basic components are shown in Figure 2-1.

The Drive has a starting operation called the "Starting Mode" of the system whereby the motor is run on resistors. With the starting operation, the motor is started in the usual manner of a wound rotor motor by means of the AC Starting Package

2-2. MAIN POWER DISCONNECT

Primary protection for the wound rotor motor and the converter is provided by a power circuit breaker. Controls are arranged to close and trip the breaker through signals fed to the Kraemer Drive. Short circuit (instantaneous overcurrent) and overload protection (time overcurrent) for both the stator and the inverter transformer and the feeder cable from the switchgear to the system trip the breaker directly.

2-3. MOTOR

The wound rotor motor is a 6 pole machine. Rated 1600 kilowatt, 6300V ac primary, FL stator, 960V on rotor at standstill, 1020A secondary full load, synchronous speed of 1000 RPM.

2-4. DIODE RECTIFIER

The function of converting the rotor ac power to dc power is performed by the diode rectifier. This **is** made possible by the ability of the diode to block voltage in one direction and pass it in the other. The diodes are arranged in a three phase bridge configuration. (Figure 2-2).

Each leg of the bridge consists of 2 series connected diodes, heatsink, associated R-C snubber circuits, and a NTVB board containing cell indicating lamps and resistor attenuators. Each diode's PRV exceeds 2.5 times the peak inverse voltage (PREV) imposed by the rotor voltage at minimum running speed (when running in KDS mode). The R-C snubber limits the recovery voltage (in conjunction with the line-to-line filter) to safe values for the cell. The attenuators in conjunction with base mounted resistors form a voltage divider to reduce ac line and dc voltages to values which can be used in the signal electronics module. A second diode bridge - minus attenuators - is used in parallel to handle the large rotor current.

2-5. REACTOR

The reactor provides three functions:

a. Limits rate of rise of fault currents so that they do not reach too high a value by the time the running contactor (MR) opens and inserts the current limiting resistor.
b. Attenuates the rotor ripple current to smooth the dc voltage.
C. As the drive increases speed, the rotor voltage drops and the inverter phases forward to maintain current. In order to produce average volts near zero the inverter must fire during the positive half cycle of the inverter transformer voltage. During this period the inverter transformer voltage will drive current through the diode bridge(s). Figure 2-3

The reactor limits this component of inverter current to ten percent of full load dc current. In the case **of** a 12 pulse circuit, two bifilar winding reactors are **used;** 1 winding for each bridge (channel) A and B.

2-6. CONTACTORS

a. SKD Contactors - The SKD contactors connect the rotor to the rectifier (diode bridges) after running speed has been reached (min. 80% speed) and the ac reistor contactors have opened.
b. Limiting Contactor (MR)
The fault current limiting contactor shorts out the fault current limiting resistor when running on the inverter. It limits fault cur-

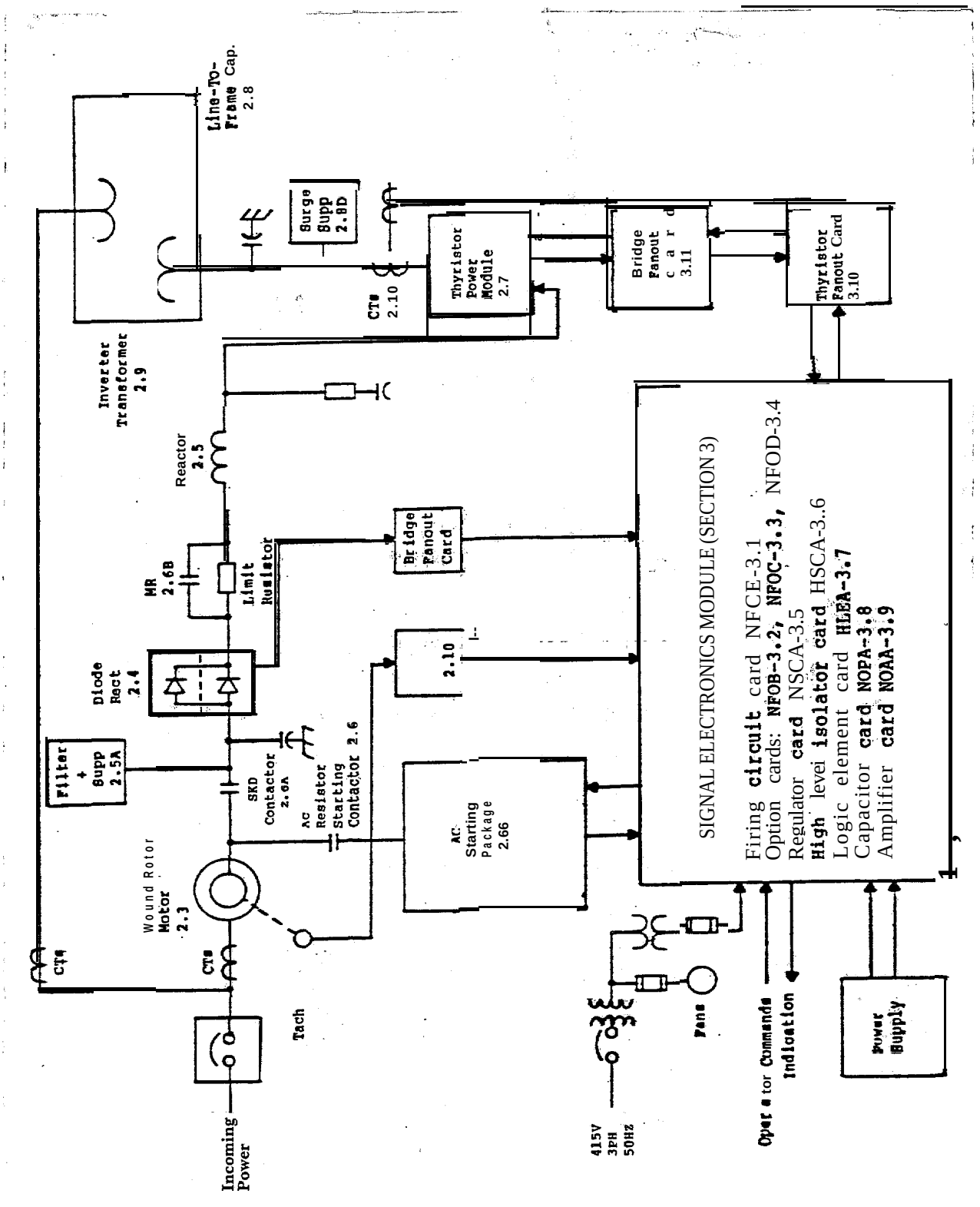


Figure 2-1. Kraemerstat Power Converter

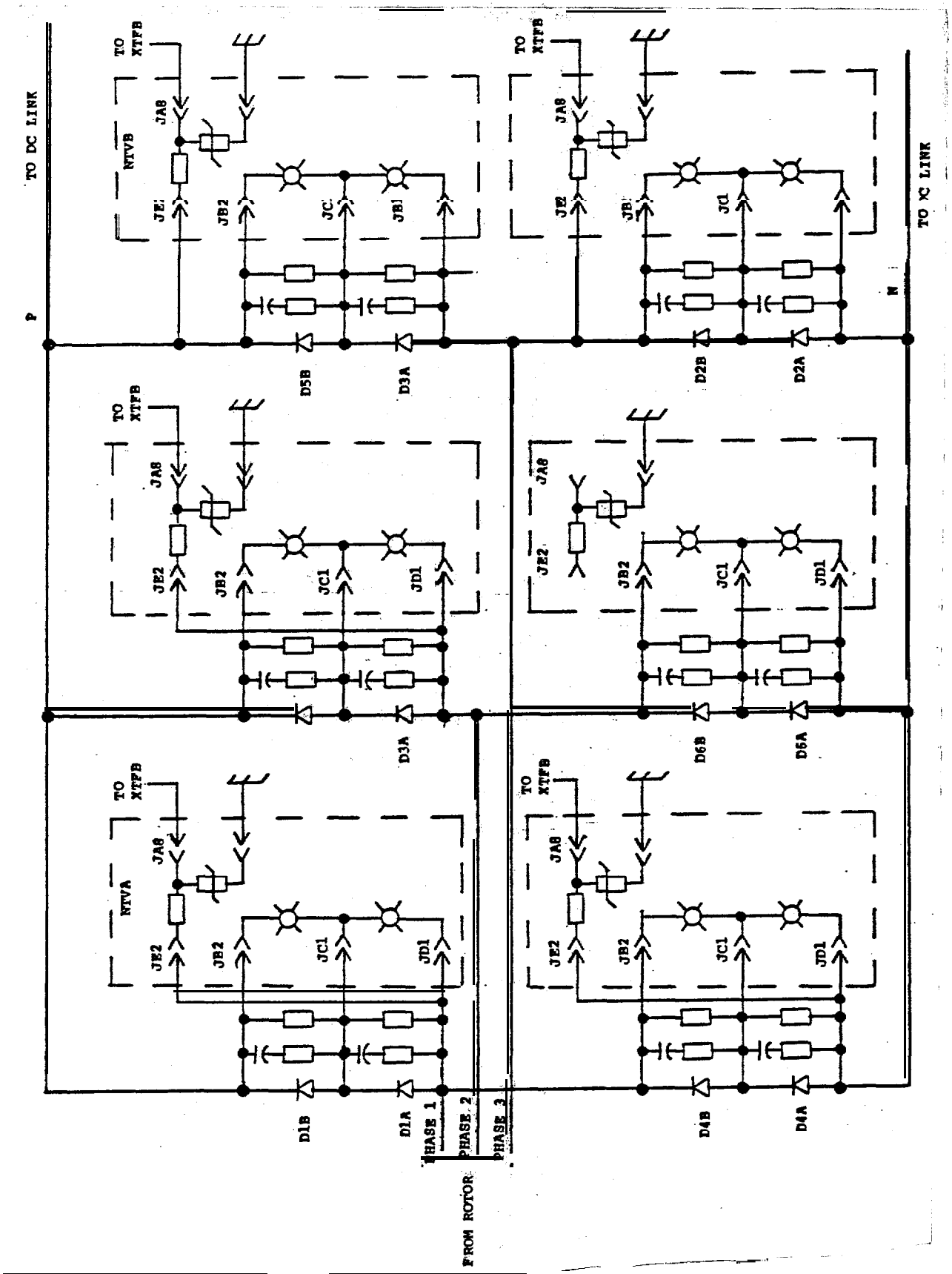


Figure 2-2. *Diode* Bridge

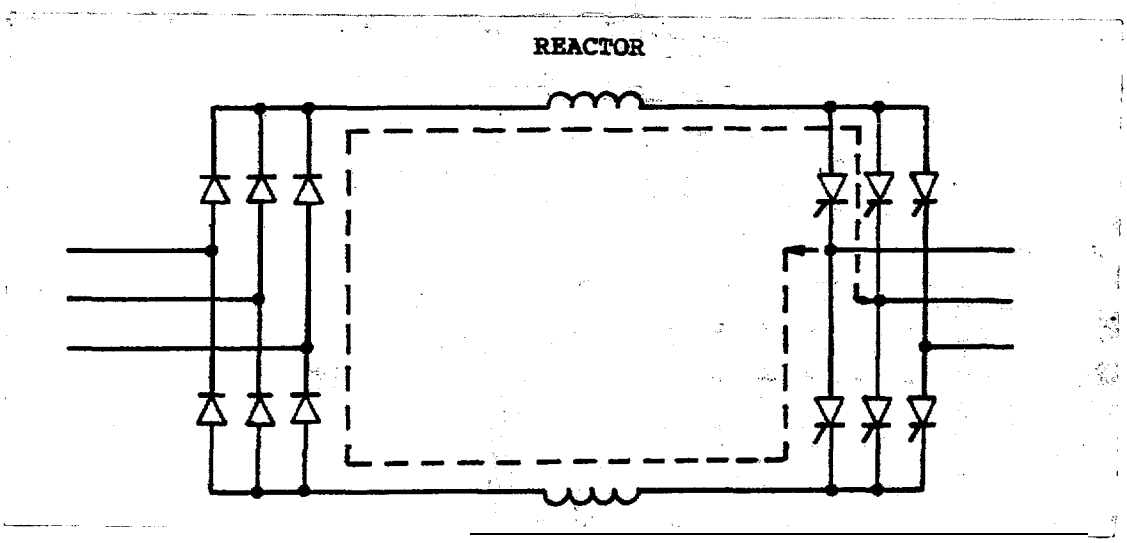


Figure 2-3. Reactor

rent by opening on faults and reinserts the resistor into the circuit.

c. AC resistor Contactors

During starting the ac resistor contactors are closed while the SKD contactors are opened. The motor accelerates through the ac starting resistors to running speed (80% speed).

2-7. THYRISTOR POWER MODULE.

The function of converting ac power to adjustable voltage dc power is performed by the converter. This is made possible by the ability of the thyristor to block voltage in both directions, while the degree of current flow in one direction is controlled by "gating" the thyristors at adjustable times on the ac voltage wave. Both the basic power circuits and the gating control circuits of the converter are modular, thereby lending the converter to the building-block principle for flexibility and versatility in meeting varied ratings and applications. The basic converter module is a three-phase bridge (Figure 2-4) consisting of six leg (L-N) modules.

A leg module for the SILCOMATIC ® Plus power converter consists of a cell assembly, a pulse amplifier and transformer (PAT) module, and a reactor. Six leg modules are combined to form a full wave bridge. Additional bridges may be paralleled to increase the load current rating of the SILCOMATIC® Plus converter. Alternately, a second bridge can be connected to second ac source, displaced 30 electrical degrees from the first ac source, to give a 12 pulse system that increases the load current rating of the converter but additionally reduces system harmonics.

A cell assembly consists of the thyristors, heat sinks, and the RC snubber circuit combined into an integral unit. The cell assembly used on these converters is called a series configuration since two thyristors are connected in series, mounted as an assembly, and share a commonleg reactor, fuse, PAT, and snubber. The snubber consists of a series reactor and an RC shunt network applied to each cell and conducting path. This arrangement protects the cells against switching surges, ringing, and excessive rates-of-rise of voltage and current during the commutation period. Without this protection, the cells would tend to conduct from changes in voltage, receive excessive changes in current, be overstressed by transient voltages, and tend to switch off after firing due to ringing. Failure would be inevitable, usually from the voltage changes.

Careful visual inspection of these circuits should be made during installation inspection and especially following failure of any cell where the cause is unknown. A cell assembly may be removed from the front of the unit by removing three bolts and unplugging the gate lead connections at the PAT module. Since the cell assembly is an integral unit, field repair should

not be attempted. All cell assemblies considered repairable should be returned to the factory for component replacement.

The PAT card located in the leg module contains two parallel circuits for amplifying the firing pulses obtained from the signal electronics and thus firing each of the two cells contained in the leg module. It also maintains electrical isolation (through the use of pulse transformers) between the signal electronics and the power circuits. A secondary function is neon cell indicating lights which normally glow when bridge power is applied but are out when the cell is shorted. Also provided is a LED indicator which lights to indicate presence of pulses to forward cell pulse transformers. Attenuator resistors are also mounted in the PAT assemblies. These resistors are connected to each of the five buses (phase 1, phase 2, phase 3, P, N), however, there is only one attenuator in each PAT. Therefore, five PATs are required to furnish all five signals. This implies that one PAT in a bridge has no attenuator connection. These attenuators are used in the signal electronics to develop phasing signals, voltage feedback, and cell voltage sensors.

2-8. FILTER CAPACITORS AND AC FILTERS

a. -Line-to-Frame Filter Capacitors. These capacitors are used to suppress voltage spikes caused by switching surges. They provide a discharge path between the secondary windings of the transformer and ground.

b. Line-to-Line A-C Filters.
These filters are designed to prevent voltage oscillations which would result from switching of the thyristors as commutation progresses from one phase leg to the next. This affect becomes more severe as phase retard increases to the point of maximum ripple (between 60" and 90" retard, depending on actual inductance of the load circuit), where the switching change of voltage approaches peak line-to-line ac volts ($2E_{L-L}$). Without these filters, overshoot would not only stress the inverse blocking capability of the cells, but ringing could occur sufficiently to switch off the cells after firing.

c. DC Filters.
DC filters, calculated for rated load and inductance values in the DC bus loop, are used for damping purposes within the dc system and to aid commutation when the dc current is discontinuous between successive conduction periods.

d. Surge Suppressor.
Surge suppressors, connected line-to-line, provide additional protection against transients caused by "chopping" excitation current on the converter transformer. Their function is to clip line-to-line surges at a value between the peak voltage level of the circuit and the PRV (peak reverse voltage) rating of the thyristors. When the spillover conduction point of the suppressor is reached, the impedance of the device changes instantaneously from a high value to a very low value. It then remains at the low value until the ac line-to-line returns to normal. No adjustments for these devices are provided or intended. They are calculated, fixed values.

2-9. INVERTER TRANSFORMER.

An isolation type transformer couples the rotor electrical power back into the main ac line through the use of thyristor firing. A thermal switch provides overtemperature detection.

2-10. CURRENT TRANSFORMERS.

Each channel of the drive has main *ac* current transformers that are applied on the thyristor ac bus for regulation and protective sensing. The secondaries of these transformers feed the primary of the auxiliary transformers. The secondary of the auxiliary transformers is rectified and fed to the burden resistors mounted on the thyristor fanout card (XTFA). The signal obtained from the burden resistor is directly proportional to the current in the dc line. This signal is used for regulator feedback as well as fault monitoring.

2-11. SPEED FEEDBACK ▪ PULSE TACH

A pulse tachometer is provided at the motor. The generator will generate 120 pulses per revolution (120kHz. at 1000 rpm) . The generated pulses are converted (via a pulse-to-analog conversion) at the SKD control so that 10Vdc = 1 000rpm for speed regulation.

2-12. SIGNAL ELECTRONIC MODULES (MEM)

The 6 pulse Static Kraemer drive has one signal electronic module that consists of all the electronic circuitry of the drive. This circuitry is packaged on 2 1 printed wiring boards, namely 5-NGPA, 5-HLEA, 2-NOAA, 2HSCA, NVAA, HRRB, NSCA, NFCE, NFOB, NFOC, NFOD. Schematic information and brief application information is available in the backsheets which can be identified as DS3800XXXX, where

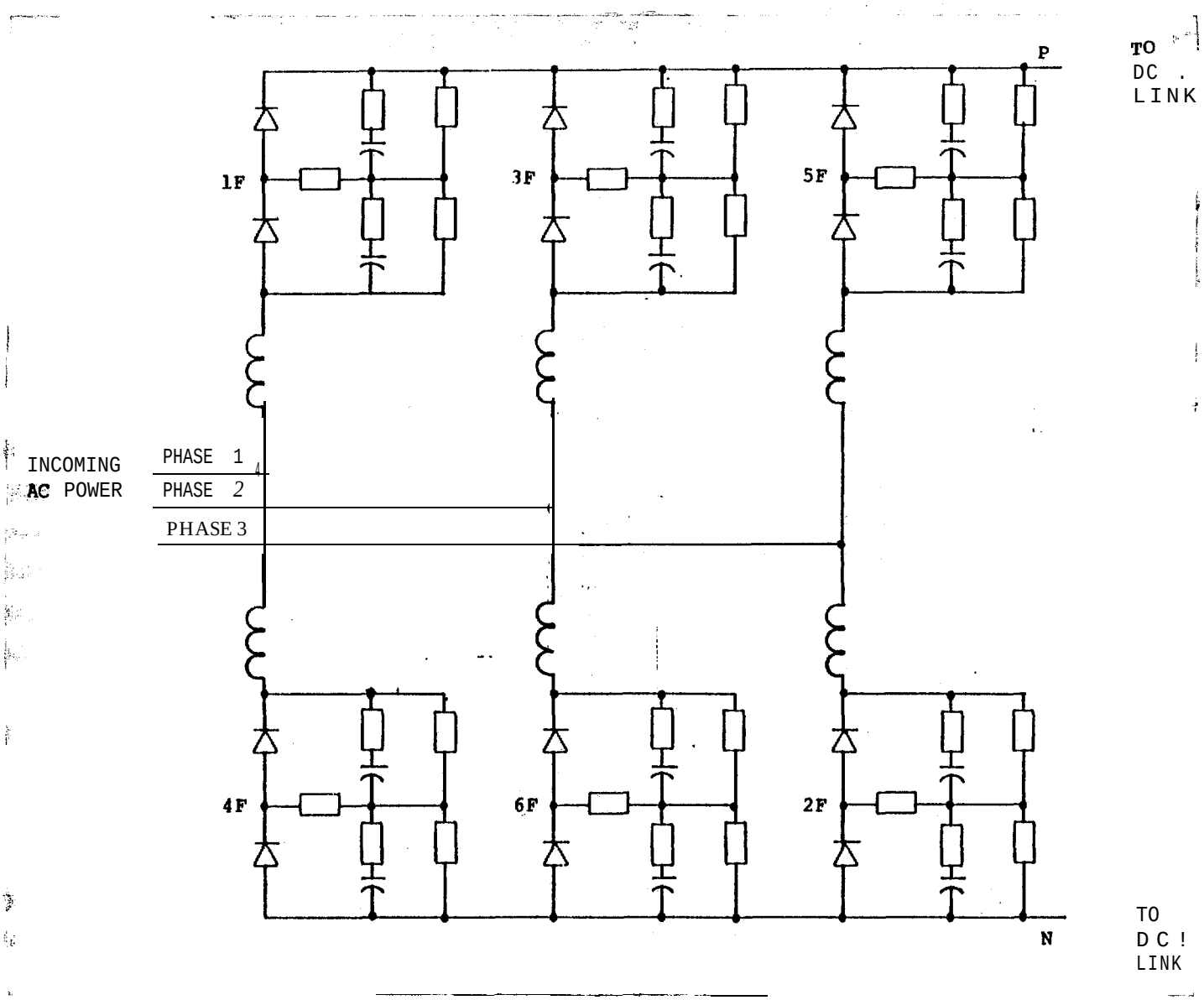


Figure 2-4. Series Cell Bridge Configuration

XXXX is the 4 letter name for that board. Brief functional descriptions are in Section 3.

2-13. SIGNAL ELECTRONIC POWER SUPPLY

This base-mounted power supply accepts 110Vac as an input and develops + 15, -15, +5 and +28 volts for use by the signal electronics modules. Circuits included are:

1. Incoming power circuit breaker.
2. Current limit protection on all buses.
3. Remote sensing on +5 volt bus.
4. Potentiometer adjustment of all buses.
5. Power supply status indication.
6. Logic input to signal electronics module - low voltage on any bus is detected.

CHAPTER 3

KRAEMERSTAT ELECTRONIC MODULE

3-1. FIRING CIRCUIT BOARD FUNCTIONS

The basic function of the firing circuit (NFCE card) is to convert the reference obtained from the drive regulator to a proportional DC motor voltage. In addition, it provides converter fault protection and regulator feedback signals.

The firing circuit on the NFCE may be divided into the following functions:

a. Phase-locked-loop
b. Biased cosine wave generation
c. Cell selection logic
d. Pulse train **logic**
e. Suicide
f. Feedback signals
g Logic overcurrent
h: Suppression
i. Overvoltage detection

a. Phase-Locked-Loop. (Line Synchronization) Bridge thyristor gate pulses must be synchronized with the ac line to achieve proper firing order and timing. The phase-locked-loop times the firing of the bridge thyristors with respect to the ac line while providing noise isolation. It isolates cell firing commands from line disturbances such as commutation notches and random spikes generated by other equipment in the plant. Line voltage (phase 1 - phase 3) is obtained using a differential operational amplifier with signals from the option board buffers. Isolation is provided by external resistive attenuators in conjunction with the differential amplifier. This voltage is then filtered (90" lag) to remove commutation notches and noise spikes with the resulting smoothed sinusoid fed to a zero crossing detector which provides a logic "1" output when line voltage is positive and logic "0" when line voltage is negative. The output is then used as the phase lock loop reference. The output of the phase lock loop is a synthesized square wave with a 50% duty cycle. This feedback waveform is fed, along with the reference wave, to a digital phase difference detector (an "Exclusive OR" element), that is, an element having two inputs which produces a "1" output whenever the inputs are different.

Note that the output of the difference detector is a pulse train whose pulse width is proportional to the difference in phase between the reference and feedback signals. This error signal is applied to an integrating operational amplifier. The result is that its output voltage increases when the reference signal leads the feedback and decreases when the feedback signal lags the reference signal (auxiliary logic steers the error signal to the integrator + and - inputs). The output of the integrator drives a voltage-controlled oscillator which operates at 768 times line frequency at steady-state. The oscillator output is then fed to a binary counter which, along with auxiliary logic, produces a digital signal which is a

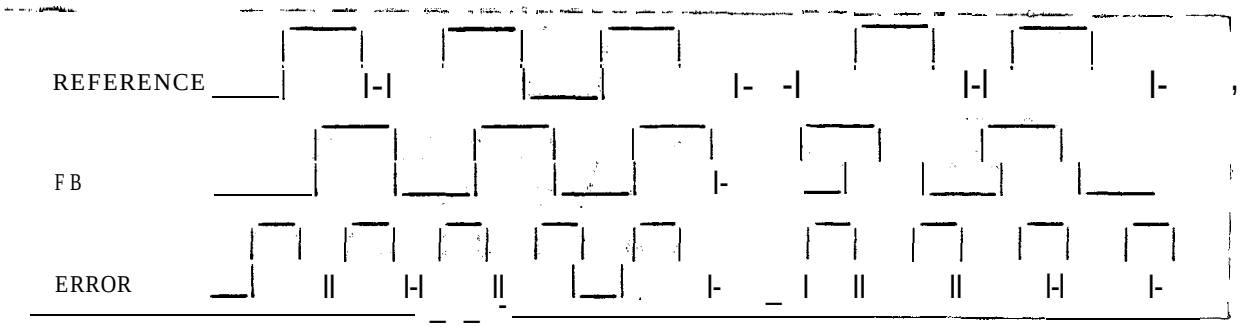


Figure 3-0.

logic "1" for 384 counts and logic "0" for 384 counts repetitively. This signal is the feedback previously described. Many other oscillator outputs are available. By using these outputs, signals shifted 15, 30, 60 degrees, etc. are obtained. These are used for timing the firing signals. An independent phase-lock monitor operates a card mounted (NFCE) indicating light ("PLL") (normally on). The light goes off to indicate when a differential of 30 degrees or more is sensed.

b. Biased Cosine/Reference Comparison. This produces a linear converter transfer function from analog reference to converter output. An analog cosine wave is generated. The gating signal is then generated by comparing the cosine wave to the analog reference from the drive regulator. Once the fire command signal is generated, control logic increments the cosine wave 60' (lagging) for comparison with the reference so that a fire command signal for the next cell may be generated. In this way, the same cosine wave and comparator may be used to generate firing commands for all cells.

c. Cell Selection and Control. Basic firing circuit control logic is stored in a digital read-only memory. This logic keeps track of the cell being fired and also selects the next cell to be fired. It also implements inverter limit and several force-fire modes to aid in controlling fault conditions.

d. Pulse Train Firing. Improves drive light load operation (discontinuous current operation). It assures that a cell which is turned off by instantaneous changes in line voltage will be refired immediately. Power Darlington transistors are used in the final output circuit to drive the pulse amplifiers, located in the thyristor bridges. Darlingtons have the advantage of high speed, necessary for transmitting the pulse train firing pulses to the pulse amplifiers. The actual pulse train is produced with a period of 4.5 microseconds and a 30% duty cycle. The fire command is used to initiate the circuit so that no partial pulses are produced. Information is furnished to route the pulse train to the correct cell pair. Also included in the driver circuitry are inhibit signals to prevent the generation of stray firing signals during power up or down of the power converter.

e. Suicide. A logic signal input will cause the firing circuit to go to a suicide condition; that is, no firing pulses will be generated. If a suicide signal is applied with the drive running, firing pulses will continue to be generated at retard limit until the current goes to zero.

f. Feedback Signals. The source of the current feedback is the burden resistor on the output of a diode bridge connected to the AC current transformers. The scaling of this signal is 7.5 volts at 150% current. Voltage feedback is derived using signals from the option card buffers. A differential amplifier measures volts from P to N. The differential OP amp output is then scaled with voltage feedback (VFB) pot with the resultant signal available for regulator use. This is normally adjusted to 4V for rated dc voltage.

g. Logic Overcurrent. Logic overcurrent (LOC) may be thought of as a kind of digital current limit. An analog comparator continuously monitors the burden resistor current feedback. When a preset current level is reached, the output of the comparator causes the firing angle to go **to** the retard limit, which should cause the current to decrease below the comparator preset. If the reference again increases the current above the preset value, the process will repeat with the current being regulated to approximately the preset value. The LOC setting is tied to the suppression setting, that is, logic overcurrent operation occurs at 80% of the suppression value. Normally suppression will be set at 1.4 times the highest power converter short time rating. This results in a LOC setting at 1.1 times the highest short time rating.

h. Suppression. An analog comparator monitors the current feedback signal obtained from the burden resistor. When current exceeds 1.4 X the highest short time rating value, a flip-flop is latched, sealing in the fault condition and turning on the "SUP" indicating LED. The firing circuit logic signal forces the firing angle to the retard limit under the constraints of the inverter limit circuitry, until zero current is attained. The forced firing is released and drive reverts to normal operation. If a second suppression occurs within 2 seconds of the first, after zero current is attained, the firing pulses are removed and the drive is transferred to standby.

i. Overvoltage Detection. Voltage feedback is fed to a comparator and compared with a fixed reference. If the peak of "VFB" is greater than 8.8 volts, 1OV becomes a logic one. This signal is latched, indicated by OV (a red LED) which becomes luminous and is transmitted to the firing circuit. The firing circuit fires the diametric cell of the positive cell selected. The signal (IDFO) is also transmitted to the fault circuit.

3-2. NFOB BUFFERS

The buffer amplifiers are used to electrically isolate and unload the attenuator signals. Attenuator signals for Phases 1, 2, and 3 and the positive (P) and negative (N) DC buses come from connections existing in the master bridge. Connections are between these buses and five out of the six pulse amplifier and transformer (PAT) cards through dropping resistors on the PAT end. These signals are then transmitted through 10-pin ribbon cables to the bridge fanout card(s) (XBFA) and onto the thyristor fanout card (XTFA). The daughter board resistors on the XTFA card scale the signals and feed them to the buffers on NFOB. The buffer outputs are then fed back to the firing circuit board (NFCC).

WARNING

Never apply bridge power with converter in test mode; the purpoose of the test oscillator is to provide a reference signal to the firing circuit phase-lock-loop so that correct converter operation may be insured without applying bridge power.

3-3. NFOC OPTION BOARD FUNCTIONS

a. Inversion Fault Detector. Commutation on a 3 phase bridge occurs between cells connected to the positive bus and between cells connected to the negative bus. Cell' one commutates to cell 3 which, in turn, commutates to cell 5. Cell 5 commutates to cell 1.

On the negative side, cell 4 commutates to cell 6, 6 to 2, and 2 to 4. See Figure 3-1.

Cells 1 and 2 conduct on PH1 to PH3 voltage. When cell 3 is fired, commutation will take place from Vl-3 to V2-3. Commutation must be complete while voltage 2-3 is more positive than l-3. See Figure 3-2.

This point occurs when Vl-3 is 300" from its positive-going zero point. If cell one is conducting after this point, commutation will not be completed and an inversion fault exists.

The 3 phases and the positive and negative bus voltages to neutral are obtained from the NFOB board. Each phase is compared to the positive voltage and the negative voltage separately. Figure 3-3 shows the circuit for cell #1, and the voltage wave shapes for PBUS-N and the phase-N voltages.

Voltage wave shapes POS BUS-N and phase voltages = 150" (wave shapes caused by cells 2-4 and 6 omitted for clarity). While cell one is conducting (= 150 to = 270 + COM), the PBUS-N and phase 1-N voltages are the same, The P5 bias on the inverting input of the comparator will drive the comparator output to -15V. Point A will be a

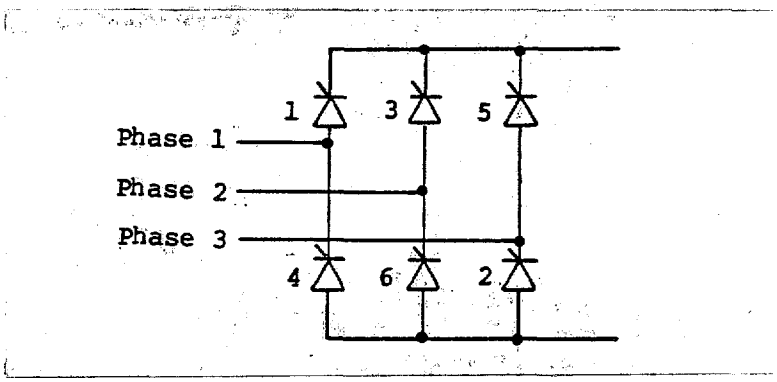


Figure 3-l. SCR Bridge

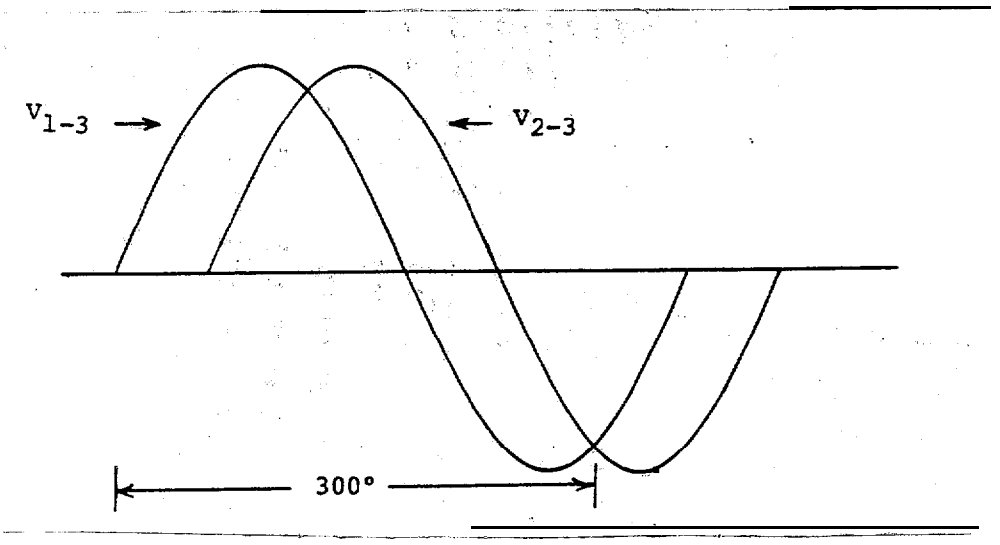


Figure 3-2. Voltage Phasing

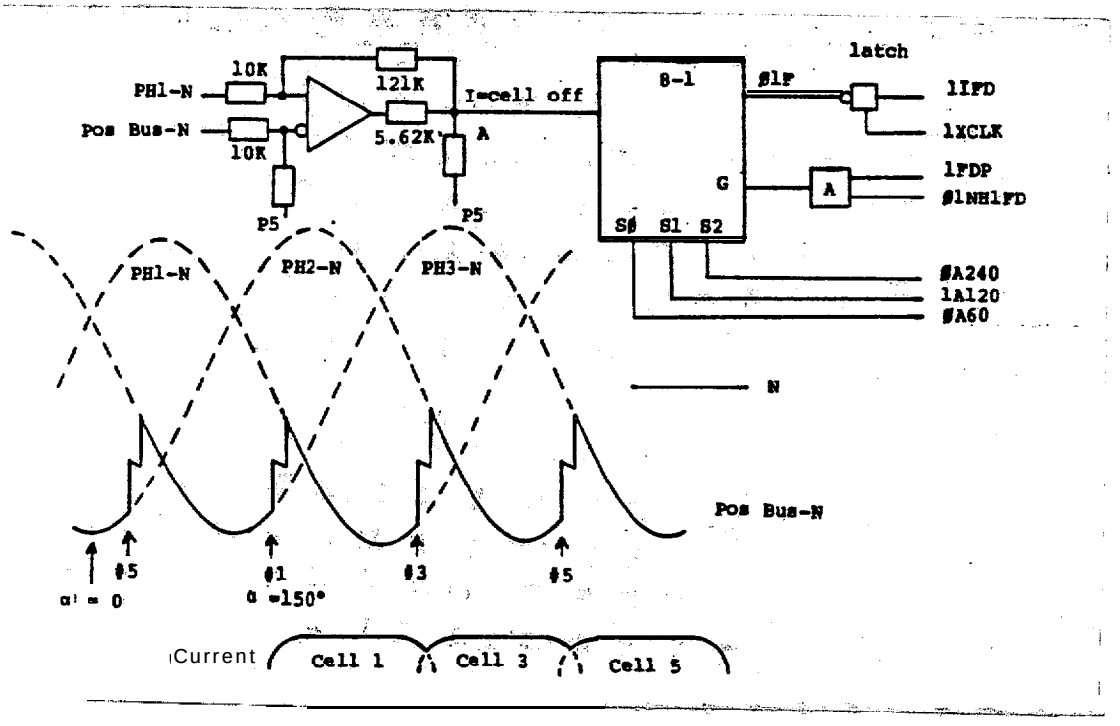


Figure 3-3.

logic 0. If commutation is complete by = 300" line 1-N will become more positive than PBUS-N. The output of the comparator will be driven to positive 15V. Point "A" will be a logic 1. The outputs of the comparators are connected to a data select multiplexer which selects each cell during its = 300 to 360". Information for the selection is obtained from the NFCE. Figure 3-4 shows the wave shapes of these signals. If cell one is conducting during the 300" ▪ 360" period, a "zero" will be transmitted to the multiplexer output providing the gate is a zero. This sets the latch in conjunction

with the NFCB clock pulse. 1IFD becomes a logic one. 1FD is fed to the NFCC which omits pulse 4 and advances 60" to pulse 5 in an attempt to pull out of the fault.

The NFCE sends a "1" signal (1IFDP) to the logic element feeding the multiplexer gate starting 15" after Alpha equal to 300" lasting for 15". This permits the multiplexer to transmit the selected cell state to the latch, pro vided the firing circuit has detected presence of current (IONHIFD = 1). When IFDP returns to a zero, it resets the latch by taking the gate high in preparation for the next cell firing. IFD is also fed to the fault circuit (see P7-6E2).

Multiplexer Function Table

S2	S1	SO	G	Output	Cell
X	X	X	H	Z	X
L	L	L	L	O	1
L	L	H	L	1	6
L	H	L	L	2	Not Used
L	H	H	L	3	Not Used
H	L	L	L	4	3
H	L	H	L	5	2
H	H	H	L	6	5
H	H	H	L	' 7	4

Table 3-4

b. Thermal Model. The purpose of the thermal model is to protect the power converter against long-time cyclic-type overloads. Whereas the suppression circuitry monitors the instantaneous value of current, the thermal model observes the long time (5-120 seconds) heating effects of the load current.

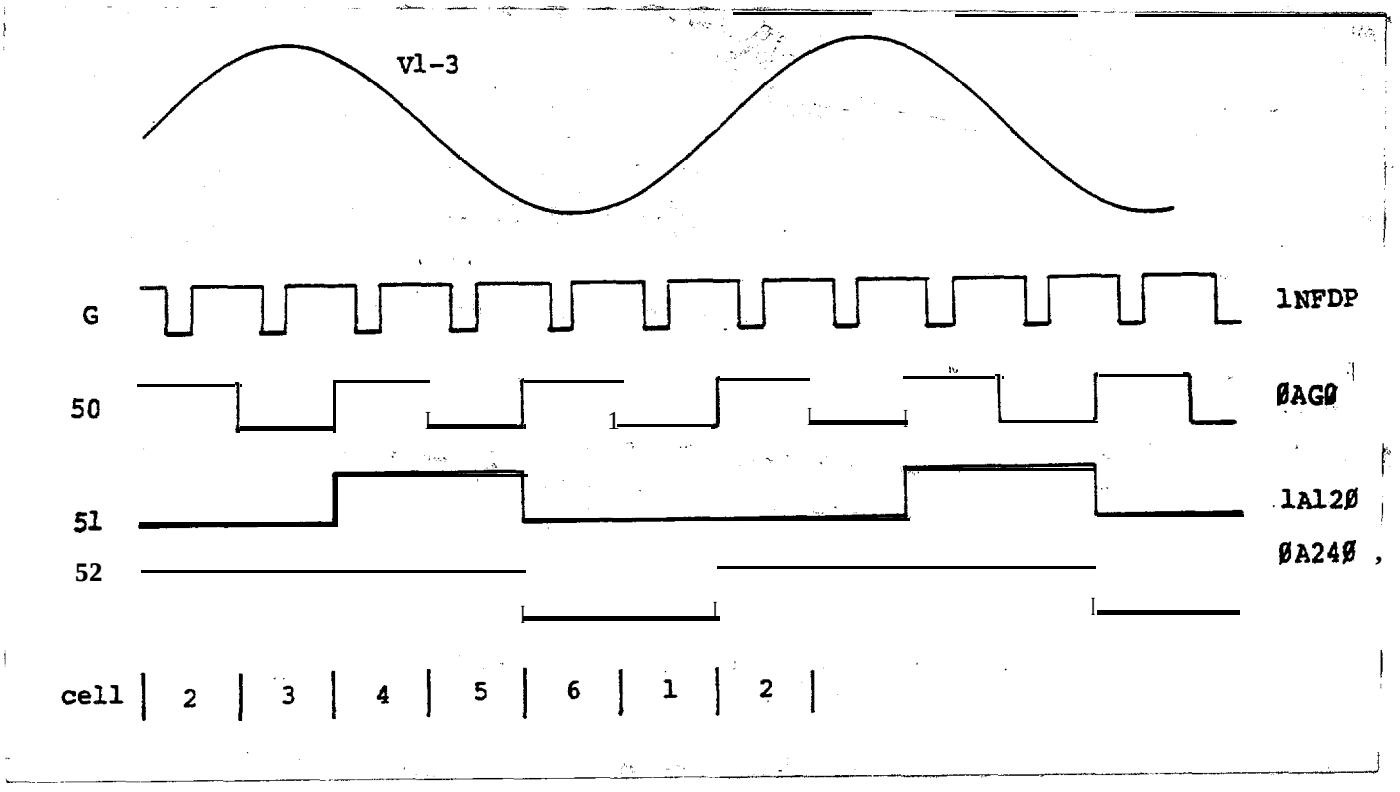


Figure 3-4.

The thermal models are pre-conditioned to rated ambient temperature. Ambient temperature is defined as the converter-rated ambient temperature. This **is** usually either 40°C or 50°C. An analog computer-like model simulates the thermal characteristics of the thyristor and heatsink. Using the current feedback signal (CFB) as an input, the model calculates the instantaneous cell junction temperature. The output corresponding to the temperature of the forward thermal model is compared with a fixed 110°C reference. When a value of junction temperature exceeding the reference temperature is calculated, the CTMP lamp on the NFOB comes on; the same signal **is** ANDed with the fan loss signal and fed into a timer. The output of the timer, after a time delay depending on 0ZCXD1, is latched and fed into the fault signal. In order to fully use the cell capabilities, the ambient input to the thermal model is adjusted to 15 below the actual ambient temperature (in this case ambient temperature is 40°C) so that CTMP will indicate a cell temperature of 125°C (125°C represents the highest "safe" junction temperature). The timer is made practically instantaneous by connecting OZCD 1 to the NFCC timer. The latch is reset by the NFCC suppression reset switch.

3-4. NFOD OPTION BOARD FUNCTIONS

a. AC Voltage Compensation. The three phase voltages obtained from the NFOB board are combined to produce line voltages 1-2, 2-3, and 3-I. The negative portion of these voltages is eliminated. The resultant positive halfs are added through a suitable gain to produce a bias voltage VCOMP. This voltage, which will increase as line voltage increases, is fed into the bias of the NFCE. An increase in bias phases the rectifier off to maintain constant output voltage.

b. Phase Loss/Sequence. The purpose of the phase loss/phase sequence detector circuit is to monitor the 3 phase voltages applied to the bridge for proper l-2-3 phase sequence and to check for single phasing. If an improper phase condition is sensed, the fault signal causes the firing circuit to be suicided. The fault is indicated by "PH SEQ" LED on the NFOD being on.

c. AC Fault Logic. The purpose of this function is to provide a means of opening the switchgear feeding the Static Kraemer Drive in the event ac fault current is not interrupted by the suppression circuit. If an overcurrent persists for longer than 27 milliseconds an output 1ACFLT **is** sent to the fault logic circuit (PAR 1-6E2) which will open the primary feeder.

3-5. NSCA FUNCTIONS

a. Speed Regulator. The speed regulator **is** adjusted for a crossover of about 2.5 radians. A lead in the speed feedback (C12A, 12B X R20) with C12A + C12B = XX MFD and R20 fixed at 100K gives good response. This lead is compensated by a lead in the feedback of the FWD gain consisting of Cl 6 (R54 + 47.5K). Both of these are set at about 2.5 radians at 60 Hz., (2.0 radians at **50** Hz.)

$$\frac{1}{C16 \times (R54+47.5K)} = 1.96 \text{ and } \frac{1}{C12 \times 100K} = 2 \text{ radians}$$

The gain in the circuit is set by the value of Cl6 and may be adjusted by R3. With R3 set in the center, the crossover of the speed regulator should be about 2.5 (2.0) radians. The response to a small step change in speed reference should be nearly critically damped with a 0.4 (0.5) second response.

b. Current limit on the inverter is set with R4 on the NSCA board and is set at 7.5 volts corresponding to 150% rated current.

c. Logic is provided on the NSCA to suicide the speed regulator and firing circuit from signals during operation.

1. Pin 80 SUAX 1 suicides speed regulator - not used
2. Pin 74 **SUA** 1 suicides speed regulator and firing circuit 0 unsuicides firing circuit and permissive to unsuicide speed regulator
3. Pin 76 OSUS 0 suicides speed regulator 1 suicides S.R. if SUA = 0

Further logic is provided to accommodate testing.

1. Pin **67** OSUA 0 suicides speed regulator and firing circuit
2. Pin 66 ORLA 0 unsuicides all regulators
3. Pin 64 ORLT 0 unsuicides firing circuit
4. Pin 59 OSUSX 0 unsuicides firing circuit

d. A filter for voltage feedback is provided to remove high frequency ripple.

3-6. HIGH LEVEL ISOLATOR CARD HSCA.

Remote and local switches are connected to the input driven by 115Vac. Light isolation is provided and inverts the input (a closed contact gives a logic 0).

3-7. LOGIC ELEMENT CARDS HLEA.

1. Six AND/OR gates are provided on each card. A true and not true output are provided.
2. Two four-input AND gates are provided per card.
3. Three inverter gates are provided per card.
4. Three relay drivers per card are provided. A logic zero provides 28V to drive a relay. An inverting logic output is also provided.

3-8. GAIN PROGRAMMING CARD NGPA.

The following functions are provided:

1. Two general purpose operational amplifiers.
2. Two three-input NAND gates.
3. Two analog (FET) switches, a high logic of 2 volts closes the switch. A low logic of 0.8V opens the switch. Input current is 0.8 mA for a low and 40 microamperes for a high (2 low power Schottky loads). The switch is good for 60 MA continuous current. Care should be exercised when working with the switches since they are electrostatic sensitive and can be damaged by voltages in excess of ± 15 volts with respect to ACOM.
4. Two general purpose dual comparators. The input voltage is compared to an adjustable reference. When the input is less positive than the reference and more positive than the inverted reference, a logic "0" is obtained on the non-inverting output. When the input is more positive than the reference or more negative than the inverted reference a logic one output is obtained on the non-inverting output. One comparator provides separate inputs for positive and negative voltages.
5. One relay driver which provides 28V at 350 mA is provided. A logic zero on the input activates the output.
6. One time delay circuit. When both inputs (or the last of either) go from a low to a high, the non-inverting output changes from low to high after a time delay. The time delay is adjustable from O-10 seconds.
7. Analog Multiplier/Divider. The two analog inputs PX-NX and PY-NY are multiplied together and appear at the VO and NZ outputs. The input and output ranges are ± 10V full scale. Inputs above 12V may damage the chip. The gain expression for the multiplier is:

$$VO\ (\text{to COM}) = \frac{(PXIN-NXIN)\ (PYIN-NYIN)}{10} + NZIN$$

Basic accuracy is up to 40MV constant output offset at 25°C. The output offset remains constant regardless of the input.

A daughter board is provided for variable components.

3-9. AMPLIFIER CARDS NOAA.

Each card contains the following components:

1. 4 inverting general purpose operational amplifiers
2. 2 differential operational amplifiers
3. An adjustable clamp for use with (1)
4. A daughter board for variable components.

3-10. THYRISTOR FAN-OUT BOARD (XTFA)

The XTFA board interfaces the power bridges to the signal electronic module. The board is base-mounted in-the control compartment. Inputs and outputs are as follows:

Connector Designation	Cable/Signal Description
JA	34-pin ribbon cable to the front of the firing circuit board (NFCE). This cable carries firing pulses from the NFCE and phasing signals to NFCE.
JB	10-pin ribbon cable to the XBFA card, mounted over the bridge compartments. It carries attenuator signals from the thyristor bridge.

Connector Designation	Cable/Signal Description
JC	26-pin ribbon cable to the XBFA board, mounted over the bridge compartment. It carries 28 Vdc power and firing signals to the bridge-mounted pulse amplifier.
JD	Not used here.
JE	26-pin ribbon cable to the front of the firing circuit option card (NFOB). It carries attenuator signals to the NFOB and buffered outputs back to NFCE.
JF	26-pin ribbon cable connector, used for the test mode only. To initiate the test mode, the 26-pin ribbon cable from the NFOB card is removed from JE and inserted in JF.
JC	1 O-pin discrete wire connector from the signal electronics power supply. It carries 28V and DCOM. The power is then distributed to the pulse transformers.
JH	6-pin discrete wire connector from auxiliary current transformers. A diode bridge on the XTFA card rectifies the CT signal. The CT burden resistors are also mounted on the XTFA daughter board.

The attenuator scaling resistors function as a signal distribution center. Selectable resistors are located on the XTFA daughter board and are factory selected for proper attenuation for the rated voltage.

3-11. BRIDGE FAN-OUT BOARD (XBFA)

There are two XBFA boards. One is for the diode bridge and the other is for the SCR bridge. The one for the diode bridge carries the attenuator connections to the signal electronic module by means of the ribbon cable. The one for the thyristor bridge receives the 28 volt power and firing signals from the XTFA board through a 26-pin ribbon connector. Included are 10-pin ribbon cables used to distribute the firing pulses to the pulse amplifiers. One 10-pin ribbon cable per pulse amplifier is used. Another 10-pin ribbon connector (JH) carries the attenuator connections to the XTFA board.

CHAPTER 4

OPERATIONAL SEQUENCE OF STARTING, TRANSFERRING AND STOPPING

4-1. GENERAL

The drive has two modes of operation : AC resistor run or Static Kraemer run. The drive must start in ac resistor run by means of the ac resistor package. At time of transfer from ac resistor run to SKD run the motor speed must be at least at 50% speed. After the transfer the minimum speed that the motor can be operated in SKD mode is 50% speed. The SKD would follow the local speed reference when it is selected in LOCAL mode. When in REMOTE mode SKD would follow the 4-20 mA speed demand signal.

4-2. START SEQUENCING - AC RESISTOR SELECTED

a. Initial Condition include:

1. Control power to CBCP, CBSKD and CBBF breaker at SKD control is on so that 115 volt, control power to the control relays, is energized.
2. Relay ACRC picks up so that the ac resistor contactors are closed.
3. Main breaker is at normal (NO LOCKOUT).
4. Run permissive is present.

b. Initiate a start command so that the START contact to the SKD control closes momentary, main breaker closes and motor accelerates by means of the ac resistor package.

c. The motor speed can be changed by varying the ac resistor package resistance. For detail speed control in ac resistor run please consult the instruction section of the ac resistor package.

4-3. STOPPING FROM AC RESISTOR RUN

a. Initiate a stop command so that the stop contact to the SKD control opens momentary.

b. Main breaker opens, ac resistor contactors remain closed. Motor speed coast down to zero.

4-4. SKD MODE RUN

a. Initial condition include :

1. Control power to SKD is on .
2. SKD cooling blowers which providing cooling to the thyristor are running.
3. SKD is ready, i.e. main breaker is open, no fault is detected by the SKD control, the drive is not in test mode, main breaker is at normal (NO LOCKOUT), and SKD is selected. The ready light at SKD control is on.
4. Starting and run permissive are present.
5. Relay ACRC picks up SO that ac resistor contactors are closed.
6. Current limiting contactor (MR) in SKD control is open.

b. Initiate a start command so that the START contact to the SKD control is closed momentary.

c. Relay MBKC picks up and closes the main breaker. Current limiting contactor (MR) picks up bypassing the current limiting resistor.

d. Motor accelerates, Ready light on SKD control panel goes out.

e. When the speed reaches to 55%, which is detected by the SKD control, the SKD commands to close the SKD contactors.

f. After the SKD contactors are closed, the SKD control will pre-condition to ask for 20% current.

g. When the current reaches 15%, the control commands the ac resistor contactors to open. SKD transits to speed regulating mode. The motor speed will follow the speed demand signal.

h. While in SKD mode, the control will clamp the speed to minimum of 50%. If for any reason the speed drops below 50% then the control will declare fault and opens the main breaker.

i. The speed range in SKD mode is 50 to 98.5% rpm.

4-5. STOPPING AT SKD MODE

a. Initiate a stop command while the drive is in SKD mode.

b. The main breaker opens, MR contactor open, and SKD contactors opens, the AC resistor contactors close.

c. Motor coasts down.

CHAPTER 5

INSTALLATION GUIDELINES

5-1. GENERAL REQUIREMENTS FOR RECEIVING, HANDLING AND STORING EQUIPMENT

a. Receiving Equipment

All items of equipment, before leaving the factory, are carefully inspected and packed by personnel experienced in the proper handling and packing of electrical equipment. Upon receipt of any apparatus, an immediate inspection should be made for any damage sustained while enroute. If damage is evident or an indication of rough handling is visible, a claim for damage should be filed with the carrier at once. Both the transportation company and General Electric Company should be notified promptly. Information as to damaged parts, part number, case number, requisition number, etc., should accompany the claim.

It is recommended that the equipment be placed as close as possible to its permanent location, or at its storage location, before unpacking. Unpacking should be done with care to prevent parts being damaged or marred,

Some parts, such as bolts, screws, etc., are packed in special containers to keep them together until it is ascertained that all parts have been removed from the packing material and quantities checked.

Indoor equipment should be placed under adequate cover immediately upon receipt because packing coverings are not suitable for out-of-doors or unprotected storage.

The equipment may be stored in the same environment in which it was designed to operate. Equipment built for operation in an outdoor environment may be stored outdoors and equipment built for operation in an indoor environment must be stored indoors. (Refer to storage section for additional storage preparation.)

Any items removed from the equipment for storage preparation (or any other reason) must be replaced prior to placing the equipment in service.

b. Handling

Equipment is generally shipped by truck. Small parts or modules may be shipped by air. Vibrations and shock inherent in rail shipment make it the least preferred shipping method.

Refer to equipment outline drawings for preferred methods of lifting and moving the equipment.

The equipment's center-of-gravity should always be carefully observed so that the equipment will not fall.

The limit of shock to which any equipment should be subjected must not exceed an instantaneous force of 3 times the equipment's weight (3 g's for 11 milliseconds maximum).

The limit of vibration to which any equipment should be subjected must not exceed 1 times the equipment's weight (1 g.) at a frequency rate of 10-120 Hz.

c. Only authorized personnel should handle and prepare the equipment for storage. All equipment should be secured against entry by unauthorized personnel.

If any part of the equipment is not to be installed immediately, it must be stored in a clean, dry place and protected from sudden changes in temperature and humidity. If the temperature of the storage room varies to such an extent that equipment surfaces are exposed to sweating or freezing conditions, it must be protected by a safe, reliable heating/cooling system to minimize corrosion by gaseous contaminants (activated by moisture). The internal temperature of the equipment should be kept 5°C to 10°C above that of the storage room or outside temperature (ambient temperature) to

eliminate condensation inside the equipment.

It is recommended that for maximum electronic component life that the internal temperature be maintained between + 5°C and +45°C. In no instance should the internal temperature of equipment exceed -40°C and +55°C (the maximum temperature of equipment containing CO, bottles is + 60°C).

Equipment may be stored in environments of 5% to 95% relative humidity with the temperature conditions as stated above to prevent condensation.

To minimize the accumulation of particulate contaminants, a breathable type cover (canvas, etc.) should be used to cover the equipment while in storage.

DO NOT USE plastic type covers (polyethylene, etc.) without specific provisions to prevent condensation beneath this type cover.

For outdoor equipment storage, an inspection should be made to insure that all container openings are sealed and closed against wind, rain, snow, sleet, flooding, dust, etc.

Heaters, air conditioners and/or air purging systems may be required to meet the above storage criteria. If so, it will be necessary to install and check this equipment prior to its operation.

CAUTION

Remove all cartons and other miscellaneous material packed inside units before using heaters to avoid afire hazard

If equipment has been exposed to moisture, it should be tested and checked by qualified personnel of General Electric Company before being placed into service.

Equipment such as batteries, air conditioners, purge systems, etc. must be inspected and serviced as required by the vendor. Batteries must be recharged every 90 days. The control should be inspected monthly for damage or signs of deterioration and corrective action taken where necessary.

Important parts such as jack screws, gears and chains of circuit breaker mechanisms, linkages and moving machine-finished parts must be prepared for storage as specified in each individual device instruction book.

Low voltage ac drawout breakers (up to 600V) remain in the equipment for storage. Low voltage dc drawout breakers (AG-7 type) may be prepared for storage separately. Medium voltage drawout breakers (5kV to 15kV) should also be prepared for storage separately. Refer to the appropriate circuit breaker instruction book.

Special Requirements

There are some cases where special preparation and packaging are required for storage and operation, e.g. - tropical or polar climates. These situations should be referred to General Electric Gompany for their evaluation and recommendation.

CHAPTER 6

INITIAL START-UP AND CHECKOUT PROCEDURES

6-1. OBJECTIVE

The objective of this chapter is to serve as a guide for inspecting and testing the SILCOMATIC Plus power converter.

6-2. PRECAUTIONS

The following precautionary advice is included to forewarn the tester against procedures which might nullify important protective functions designed into the system.

a. Do not, under any circumstances, remove gating signals from only a portion of the bridges paralleled to share the load. The converter is designed to withstand all fault conditions, provided the full complement of bridges is gated and all protective functions are operable.

b. Do not megger electronic control modules and circuits in which solid state electronic devices are present. They can be easily damaged. When meggering converter power circuits, short all buses together to avoid inducing damaging potentials across rectifying devices unnecessarily. Hi-potting the rectifier circuits is not advisable or necessary.

c. Use extreme care and caution when working with or around the rectifier circuits and modules. They are, by nature, high capacity, low impedance paths, and destructive arcing faults could occur by exercising careless procedures.

d. Do not attempt operation of the converter using open or ineffective ac current transformers.

e. Do not remove or insert printed circuit cards unless signal electronics power is off.

f. Do not apply main power to the equipment unless the equipment frame is grounded.

g. Do not operate radio transmitters within 10 feet of an operating SILCOMATIC Plus power converter particularly if the doors are open. The radio frequency (RF) signals generated by the transmitter could induce noise into the unshielded wires within the converter and create erroneous fault conditions.

6-3. INSPECTION APPLICABLE TO NEW POWER CONVERTER EQUIPMENT

a. Inspect for proper assembly and connections, both mechanical and electrical.

b. Check for loose, missing or damaged parts.

c. Perform primary breaker (ac) and converter transformer inspection and tests. Set transformer taps to provide rated input volts to the inverter of 530 volts ac for 6300 volts on the primary.

d. Perform insulation megger tests.

e. Inspect mechanical assembly for compliance with assembly diagrams.

f. Check devices having moving parts for freedom of operation, adequate clearance and proper support.

g. Check wire and cables for proper sizing and insulation (especially grounding).

h. Open front doors and remove rear panels and remove any dust, trash, loose parts or any foreign matter.

i. Clear and check fans for easy rotation (without power).

6-4. TEST EQUIPMENT AND TOOLS

a. A 2-channel oscilloscope with 3 probes (for channel 1, 2 and the external trigger) and 1 current probe. The current and trigger probes must be long enough to allow current to be checked on all cells.

b. Multimeter (preferably digital).

c. Drive Test Module (DTM) or Regulator Test Module (RTM) if available. Be familiar with the instructions for the module being used.

d. Strip chart recorder.

e. Adjustable (plus and minus) dc voltage source (0 to 14Vdc) if the DTM or RTM is not available.

f. Elementaries - system: 237B9981.

g. Check all component values in daughter board as shown in sheet DA.

6-5. TEST CONNECTIONS

Connect RTM cabling (if applicable) to receptacle JLA located at top of the signal electronics module.

6-6. TEST MODE TESTS

CAUTION

*BRIDGE POWER IS NOT REQUIRED AND SHOULD NOT BE APPLIED FOR THE FOLLOWING TESTS. WHEN THE XTFA CARD IS IN THE TEST MODE (PARAGRAPH **6-6** Cl), THE **BRIDGE** POWER IS NOT SYNCHRONIZED WITH THE FIRING CIRCUIT AND EXCESSIVE FAULT CURRENTS CAN BE PRODUCED.*

Check the wiring of the following circuits and repair any opens or shorts:

1. The overtemperature switch circuit to the transformer.
2. he remote reference circuit.
3. Remove all the fuses connected to the cooling blowers (Sh. 1AE) at this time.

A. Auxiliary Control Power

1. Insure phases 1, 2, and 3 in the incoming side of the control power switch are connected in the proper sequence.
2. Insure rear panels are in place and doors are shut on SCR panels.
3. Apply control power
4. Insure the output of transformers is correct, then turn-off power and replace the fuses to the blowers, turn on control power and insure each blower at the thyristor bridge is operating properly as to direction and air flow.

B. Electronic Power Supplies

1. Determine if power supply PSFF is installed. (See elementary sheet AG or data plate on power supply.) Turn on the power supply. Measure the following voltages on the buses at the top of the signal electronic module for the settings recommended below.

C. Test Oscillator and Gate Pulse Generation

1. On both XTFA cards, which are mounted on the control panel rear wall above the signal electronic module (M02), disconnect the ribbon connector from JE and reconnect it to plug JF. This results in the test mode configuration. Verify the LEDs on the cards indicated below.

BUS	RECOMMENDED SETTING	POWER SUPPLY POT
P5	+5 ±.1Vdc	R1 (NPPB)
P15	+15±.1Vdc	R6 (NPSK)
N15	-15±.1Vdc	R7 (NPSK)
P28	+28 ±3Vdc	Unregulated Use Berg jumper (BJ1) at NPSP to change voltage level of 28V.

LED	CARD	STATUS
TST	on NFOB(1B)	on
PLL	on NFCA)1A	off
SEQ	on NFOD(2C)	off

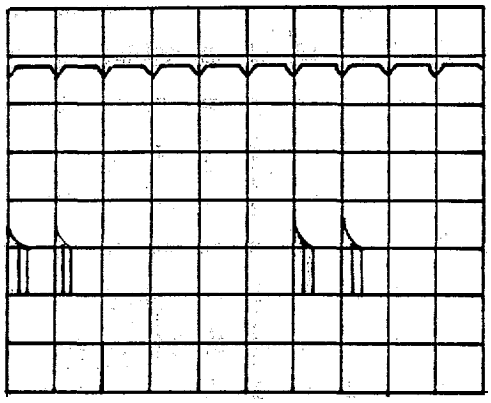
** *NOTE* **

It may be necessary to reset the NFOB cards during this test.

2. Hook up test reference(TP2) to the TREG on the NFCE card. Jumper pin 24-1E to ACOM to remove regulation reference (sh. 1B2).
3. Connect channel 1 of the scope to COS (test point 9 on the NFCE card) (sh. 1CA). Use the external scope synch connected to TP-PH2 (NFOB card) (sh. 1CD). Monitor the gate current of cell 1 on the inverter bridge (inside SCR door) with the current probe on a second channel. The current probe is connected to the white wire for the following amplitude and position checks.

4. Release the firing circuit suicide by connecting pin 64 **NSCA(lC)** to DCOM, set the test reference output for -9.0 volts. The oscilloscope presentation should appear as follows:

Channel



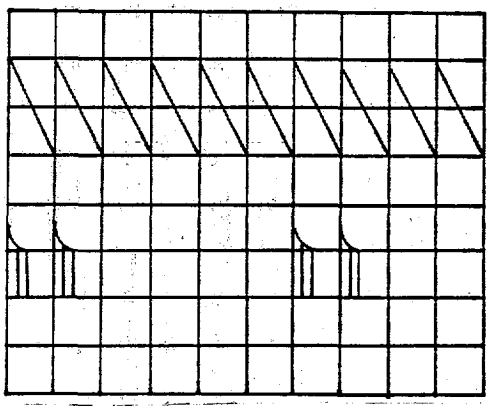
Channel 2

Horizontal: 60°/DIV
Channel 1 : 2V/DIV
Channel 2 : 0.1A/DIV

5. Decrease the test reference output to 0.0 volts and verify the COS waveform appearing on Channel 1 of the scope retards smoothly.

6. Set the test reference output to 12.0 volts to obtain the maximum retard. The scope presentation should appear as follows:

Channel 1'



Channel 2

Horizontal : 60°/DIV
Channel 1 : 2V/DIV
Channel 2 : 0.1A/DIV

7. Check the remaining forward cells in the master bridge for proper firing order and presence of pulses. As the firing sequence is followed (1,2,3,. . . 6), the firing pulses will lag the previous cells firing pulses by 60" (3.3ms on scope) (sh. 1BY).

D. Preliminary Set-Up Checks

1. Operation from local station

a. Turn off control power, select SKD mode at local station, and remove external connections from TBCH AA29C to 31C. This prevents M from closing.

b. Temporarily jumper AJ70G to AJ62G. Jumper AA29C to AC1X. Jumper AA31C to AJ76F. This will simulate main breaker (M) closing whenever the MBKC relay picks up and will simulate main breaker (M) open whenever MBKC relay drops out.

c. Connect a 0-10Vdc power supply (use RTM) to TP4-(1C) NSCA to simulate the speed feedback of the motor. (-10V = syn speed, 1000 rpm).

d. Turn on control power.

E. Fault Signal Checks

1.The following SKD faults will trip the drive without intentional time delay.

FAULT	INDICATION
a. 10% difference in speed feedback and rotor volts	HLEA - 2J CR6 on
b. 2nd suppression	NFCE - SUP on
c. Blown filter fuse	HSCA - 2M CR10 off
d. Thermal overload	NFOC - TOL on
e. Overvoltage	NFCE - OV on
f. Transformer overtemperature	HLEA - 2E CR6 on
g. KDS blower loss	HSCA - 2L CR4 off
h. AC undervoltage	HLEA - 2E CR2 on
i. Below minimum speed while running	HLEA - 2H CR5 on

FAULT	INDICATION
j. Switchgear lockout	HSCA - 2L CR10 on
k. AC fault	NFOD - ACOC on

2. The following faults will trip MR (running contactor) and reclose once, if faults persists (reoccurs on reclosure) drive will shut down.

   a. Suppression (instantaneous inverter overcurrent set at 1.4 times current limit or 2 10% full load).
   Inverter fault detector (failure of inverter to commutate).

3. With no faults and with SKD select in local mode, FAULT light will be out. Simulate each of the conditions below as follows. Verify that FAULT light comes on for each test. Most faults must be reset by the SRST on the NFCE card.

   a. Speed/Volt Differential. Connect a positive test voltage to TP3-1H (sh. 1BN) and adjust to obtain 1.5. Comparator CPC-1H will pick up. FAULT(must also ground 2J-55) light will light and latch until reset.

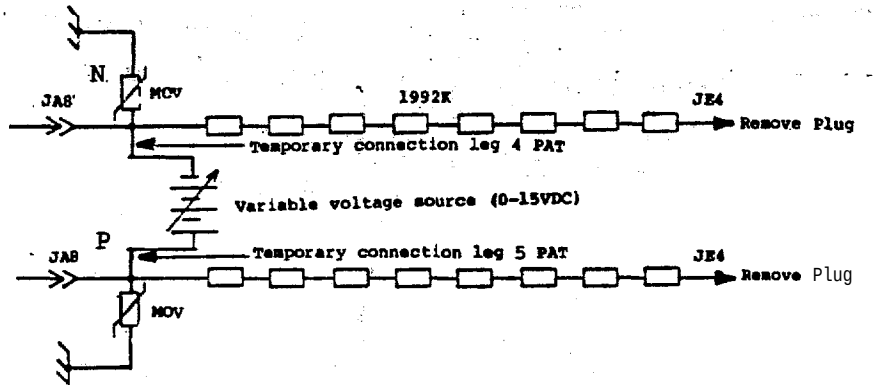
   b. Blown Control Fuse. Temporarily remove fuse FU1 (elem. Sh. AC). FAULT light will light.

   c. Thermal Overload. Connect a test reference to T1A (TP3) (elem. Sh. CB). Raise to obtain 11 volts on CFB. ILR(panel door) will light after a time delay.

   d. Transformer Overtemperature. Temporarily remove lead from the transformer overtemperature device at inverter transformer. FAULT light should light.

   e. KDS Blower Loss. Temporarily trip each blower in SKD. FAULT light should light.

   f. Overvoltage. Temporarily put drive to normal mode by moving the plug from 'JF" to "JE" on XTFA card. Turn off bridge power and control power. Remove plug JE from PAT card HPTF, leg 4 (removes NEG bus feedback). Remove plug JE from PAT card HPTF, leg 5 (removes POS bus feedback). Connect a temporary wire to ATT on both PATS as shown below. Connect a voltage source to these connections (0-15Vdc) positive on leg 5, negative on leg 4. Turn on control power only. Raise the voltage source to 15Vdc. VFB on NFCE reads 9.8V. LED OV (overvoltage) on NFCE should come on. FAULT light on door should come on.



   g. AC undervoltage, Temporarily jumper- pin 6 - 2E to ACOM, initially increase the variable voltage source to 9 volts, then decrease to 4.98 volts. Adjust R1 - 2C until fault light on door comes on. Remove jumper 6 - 2E.

   h. Suppression. Jumper pin 52 - 2C to ACOM to prevent ac fault (sh. 1CF).
   Put drive in normal mode by moving the plug from "JF" to "JE" on XTFA card. Temporarily remove R60 and R61 from daughter board of XTFA (elem. sh. 1CG). Connect a voltage source to DTFA pin 17 (sh. 1CG). Input a step voltage of 10.80 volts and immediately remove it. SUP light (NFCE) should come on, FAULT light should remain off, MR should open but M should remain closed. After 1.5 seconds, the SUP light should go out and MR should reclose. Repeat but leave the 10.80 volts on. The sequence should repeat except that after MR recloses, FAULT light and SUP light should both come on and all contactors should trip. Remove jumper at 52 - 2C.

i. Switchgear Fault. Temporarily activate lockout relay at switchgear. FAULT light should light, contactors should trip.

j. AC Fault. Temporarily remove daughter boards (DTFA) from XTFA . Connect a test reference to DTFA pin 17 (sh. 1CG) and slowly raise to 10.5V. ACOC on NFOD card location 2C should light, the fault light should light and all the contactors should trip.

k. Below minimum run. Will check later during sequencing test.

F. Thermal Model Verification

** *NOTE* **

Voltages applicable to this test appear on elementary sheet 1CE.

1. Set power supply PSFF off.
2. Connect a voltmeter (preferably digital) and a channel from the chart recorder to -TJ (test-point 6) on the NFOB card.
3. Set PSFF on.
4. Connect a test reference to test point TIA on the NFCE card.
5. Release the firing circuit suicide by connecting TP12 - 1C to ACOM (sh. 1BY).
6. Adjust the test reference applied to DTFA-17 board, so CFB (FWD) will be applied as a stepped voltage .
7. Allow test point -TJ to settle to its ambient value (TAMB) (NFOC-2B).

** *NOTE* **

This step may result in an excessive waiting period, however, the -TJ ambient value may be expedited by momentarily jumpering pin 13 of the NFOC card to either N15V or P15V on the back of the signal electronics module.

8. When test point -TJ value reaches TAMB, start the chart recorder, apply the test reference 7.5 volts as a step and observe the chart recording. At TI second, the chart recorder should indicate a change of voltage equal to THFM; if not, refer to the thermal model calibration procedures.
9. This completes the thermal model verification (sh. ICE).

G. Sequencing Checks

1. Use REF 1 of RTM input reference to pin 17 SPRF module (NSCA) to simulate speed feedback (O-10 Vdc = O-1000 rpm).
2. Make sure that start permissive, transformer overtemperature contact (if used), run permissive and stop contacts are closed.
3. Remove the burden resistors at XTFA board (removing the daughter PWB on XTFA will be same) (sh. 1CB).
4. Use REF 2 of RTM input reference to TP8-1A (NFCE) to simulate current feedback (O-5.0 Vdc = 0-100% current).
5. Turn on control power.
6. Turn off bridge power, i.e. lockout main breaker.
7. Make sure the control is at TEST mode by observing the TEST LED on NFOB card is on.
8. Jumper pin 61-2J to DCOM (pin 1) so that MBKC can operate in test mode (sh. 1BB).
9. Set REF I and REF 2 to 0.0 volt initially.
10. Initially the ac resistor contactors should be closed, the SKD contactors should be open. Control relay MBKC drops out.
11. Clear any fault detected by the drive, and select SKD mode.
12. Current limiting contactor (MR) opens, and READY light on SKD control door is on. Fault light is off.
13. Initiate a start command at the local station, relay MBKC picks up, MR closes.
14. At this moment the electronic RUN signal is on (LED CR1 of 2F is on), and firing circuit is suicided (LED CR6 of 2F is on).
15. Increase REF 1 of RTM to >5.5 volts *to* simulate speed is above 55% (LED CPA of 1G is on).
16. Relay SKDC picks up and closes the SKD contactors.
17. SKD running light is on and the control requests 20% current (LED CR5 of 2F is on).
18. Increase REF 2 of RTM to .75 volts to simulate 15% current feedback.
19. Relay ACRC drops out and opens the ac resistor contactors.
20. At this moment the drive should be in SKD speed regulating mode. Speed regulator is unsuicided (TP J of 2H goes to logic "1" state), and remove 20% current request.
21. To simulate below minimum run fault:
    Adjust REF 1 of RTM to below 5.0 volts (sh. 1BE), CR5 of 2H comes on, fault light comes on, relay MBKC, SKDC drops out, MR opens, and ac resistor contactors close,

22. Reset the fault and simulate the drive is running in SKD mode as described above.
23. To simulate stop, initiate a stop command at local station.
24. Relay MBKC opens, contactor MR opens, ac resistor contactors close, ready light comes on.
25. This completes the sequencing check.
26. Remove the jumper at pin 61-2J, remove the references at RTM and any jumper that had been used at the back plane of the module.

6-7. POWER MODE CHECKS

A. Phasing Check

1. On XTFA card disconnect the ribbon connector from plug JF (test mode) and connect to JE for normal mode.
2. Remove jumpers from AA29C and 31C and reconnect external connections. Also remove the jumper at AJ62G and AJ70G.
3. Remove main power connections from the rotor of the motor (these leads will have 960V on them during this test and must be clear of each other and ground). Can remove leads from disconnect switch.
4. Make SKD local select and it is SKD select.
5. Turn on SKD cooling blowers.
6. Remove plug JC from XTFA cards (this will prevent cells from firing).
7. Connect a test reference to TP1-1E (elem. sh. 1BZ). Adjust for 0 volts.
8. Initiate a start command to pick up MBKC (jumper run and start permissive). Main breaker (M) should close applying voltage to primary of the inverter transformer.

Verify the following LED status:

LED	LOCATION	STATUS
TST (red)	NFOB-1B	Off
SEQ (red)	NFOD-2C	Off
PLL (red)	NFCE-1A	Off

All cell voltage neon lamps at the SRC bridge should be on. Insure PH1, PH2 & PH3 phasing is 1, 2, 3(seq light should be off)

9. Set the scope to read differentially. Connect external trigger to PH2 test point NFOB-1B. Connect scope leads as indicated below and observe the wave forms shown below.
10. Observe PB-NB test points (NFOB-IB)(lCD) differentially. The waveform should remain centered at the 0 volt ref line.
11. Initiate a stop command to drop out MBKC, main breaker should trip.

B. Load Test and Current Regulator Response as a Rectifier

1. With control power off, install plug JC on the XTFA cards. All other connects SECTION 6-7 1 to 8 remain as they were. Jumper AJ56K to AJ62K to simulate SKD bus links are in. Jumper pin 28-2F(1BF) to DCOM to remove 20% current request. Jumper TP13-1C(1BY) to DCOM to unsuicide firing circuit. Make sure TP1-1E is at 0.0 volts at this time. Turn on control power(1BZ).
2. Initiate a start command to close main breaker.

   a. Main breaker closes, MR closes.
   b. Red LEDs on PATs should be lit indicating pulses are present at the cells.
   c. Firing angle should be 135".
   d. PB-NB observed differentially should indicate zero volts.

3. Observe test point CFB (NFCE- 1A) on channel 2 and VFB (NFCE-1A) on channel 1. Trigger scope from PH2 NFOB. Gradually increase reference at TP1-1E (current ref) until CFB (NFCE-1A) has about 1 volt. Wave shape should be as shown Fig. 6-7.1 : (PB-NB no longer 0).
4. Set REF at TP1-1E to 0 volts, remove jumper at TP13-1C and jumper TP14-1C to DCOM to suicide the firing circuit. Initiate stop. Main breaker trips.
5. Connect two channels of a Brush recorder to CFB and test ref input TP1-1E and follow 6.7B 1 to 4.
6. Increase the steady reference by a small step. The response of the current should be about 300 radians. (.0033 sec).
7. Repeat step 4 to open main breaker.
8. Remove all jumpers at control and terminal board.

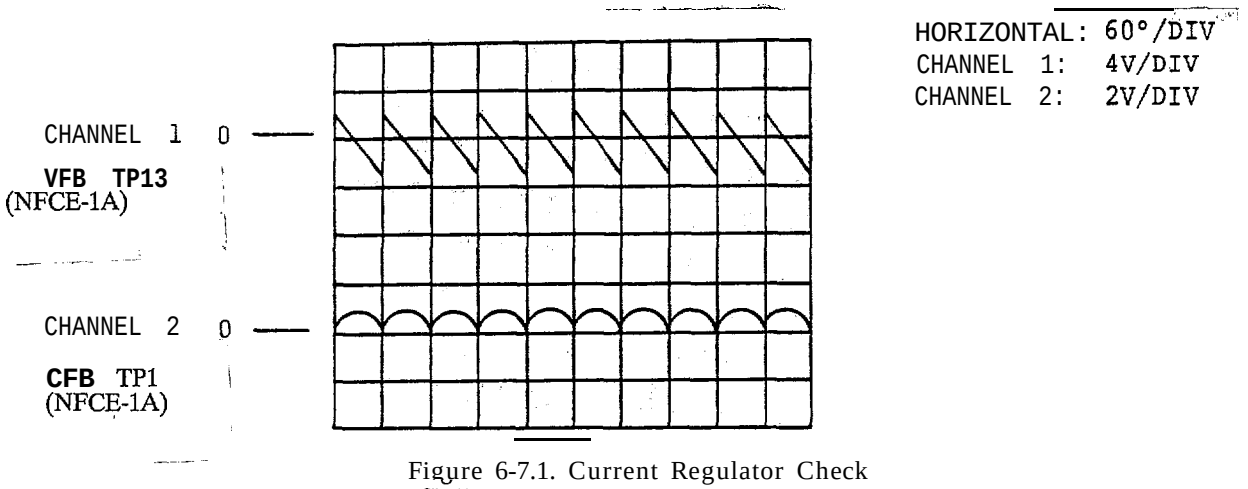


Figure 6-7.1. Current Regulator Check

6-8. POWER CHECKS AS A SKD AND SPEED REGULATOR RESPONSE

1. Reconnect the motor rotor cables, and put SKD and ac resistor contactor in normal mode.
2. Remove Brush leads and test reference into current regulator. Remove **REF to** speed feedback circuit at
3. Jumper TPI 3-1 C to release firing circuit.
   a. Connect a test reference into TP-TREG at NFCE -lA.
   b. Connect a dc voltmeter between TP15-1G (elem. sh. BT) and ACOM.
4. Initiate a start command and before motor accelerates to minimum speed, initiate a stop command to stop the motor. Observe speed feedback for proper polarity and approximately correct value.
5. Speed Regulator Response
   a. Run the motor in local mode at minimum speed, SKD mode.
   b. Connect a Brush recorder to TP15-1G (elem. sh. BT) (spd fdbk) and to TP2-1C (NSCA card) (elem. sh. BY).
   c. Apply a small step input of 0.1 volt to TP2-1C NCSA card (to increase speed). The response of the SPD FDBK should be about 2.0 Radians. (.05 sec).
   d. Remove the REF at TP2-1C.
6. Verify in local and remote mode that motor speed follows the speed demand signal.

This completes the power checks.

6-9. SEQUENCING CHECK IN POWER MODE

Refer to Chapter 4 and run the motor in SKD mode to ensure the starting and stopping sequencing is as described.

CHAPTER 7

PREVENTIVE MAINTENANCE

7-1. INSPECTION ROUTINE

A periodic routine of cleaning and inspection should be established. Such cleaning is necessary to prevent faulty operation of a device due to accumulation of dust and corrosion. Inspections should be more frequent if the demands on the equipment are exacting. Under normal conditions, the protective devices do not operate; therefore, these devices need careful inspection. If the equipment is in operation when being inspected and a short interruption of service can be permitted, the manual tripping of a protective device will afford an opportunity to observe the manner in which the equipment is functioning.

If at any time a device fails, the cause of the failure should be located and corrected before any remedy is applied and, if necessary, the part adjusted or replaced.

** *NOTE* **

When ordering parts, give a detailed description of the part and complete nameplate data on the device.
For specific maintenance information on the various devices, refer to the particular device instruction book.

CAUTION

Under no circumstances should automatically controlled equipment be started by hand until the cause of the failure to start automatically is determined and corrected.

7-2. RECORDS AND LOG BOOK

It is recommended that a regular inspection and maintenance program be employed, and that a log book be kept. This procedure will help detect malfunctioning equipment and determine the causes of the trouble. It is suggested that a form be made up and filled out at each regular inspection or when trouble is experienced. The form should contain the following data:

Normal Operation:

Operator	Converter dc amps
Date and time	Maintenance performed
Converter dc volts	Comments
Speed	

Abnormal Operation Additional Observations:

Fault lamps
Tripped breakers
Circumstances at time of trouble, such as:
Starting up Abnormal operations elsewhere but
Shutting still associated with the converter.
down
Load
Test operation
Disturbances

7-3. TROUBLESHOOTING ▪ GENERAL

A preventive maintenance program will help reduce costly downtime. Should trouble occur, the following tips will help get inoperative equipment back into service with minimum delays.

1. Become familiar with each circuit and the operation of each new component as it is installed.
2. Keep the elementary diagram handy. This is the most valuable tool the manufacturer can furnish for use in troubleshooting.
3. Keep the portable instruments handy for checking voltage, current, resistance, etc.
4. Have on hand adequate spare parts to replace failed components.

After operating properly, should the drive become inoperative, or its performance change, care should be exercised during troubleshooting in order not to get the drive out of adjustment.

Generally, the difficulty stems from a loose connection, broken wire, or failed component and can be corrected without making additional adjustment to the control. Considerable time will be saved if the difficulty can be corrected without changing the rheostat settings necessitating retuning the drive. When the system is tuned up initially, pertinent data should be recorded ▪ test point readings, etc. The following sheets give some troubleshooting hints. If a printed circuit board is changed and does not correct a trouble, replace the original card in the panel.

CAUTION

***DO NOT insert or remove a card without first turning off power. Failure to do so** may **result** in **damage to** circuitry.*

I. If the drive is shutdown and the reason for the shutdown is unknown, the following checks should be made first.

1. Check the neon cell voltage lights on the HPTF to see if voltage is present across all cells. A light which is out indicates a failed cell or a failed light.

2. Check the SUP LED on the NFCE card. If the LED is on, an overcurrent condition has been applied to the drive. Check the drive to see if there is any obvious mechanical or electrical reason for the fault. Correct any improper operating condition before depressing the reset switch on the NFCE card.

3. If there is no evidence that the shutdown was due to an overcurrent fault, the following points should be checked:

   a. Is the Phase Lock Loop (PLL) LED on? If off, the PLL is operating correctly. If the light is on, make sure that the ac power is on as evidenced by the cell voltage lights. Check that the cable from XTFA to NFCE is correctly and firmly in place. Check that the Status LED (yellow) on power supply is on. If the PLL LED is on and none of the above tests determines the cause, a new NFCE card should be tried.

   b. If the PLL LED is off, check to see if the LED indicators on the HPTF are on. If the LED indicators do not turn on, a new NFCE should be tried. If the LED indicators do turn on, the NFCE is probably operating correctly.

   c. Check the 120 volt ac control power.

   d. Check the fuses (FUSKD, FUACR) at the control power.

   e. Check the SCR gate pulse by means of a current probe. Place the drive in test mode by moving the XTFA board JE connector to receptacle JF. The purpose of the test mode is to allow gate pulse generation and converter check-out with no bridge power.

1. The RED LED on NFOB card labeled TST should be luminous.
   Jumper TP4-1E to ACOM.
2. Arrange the RTM to inject a signal into TP-TREG- 1A.
3. Monitor the gate current of SCR cell with the current probe to an oscilloscope.
4. Jumper pin 72-1C to DCOM to release firing circuit.
5. Vary the REF at TREG from -8 to + 8, at any ref level, the gate pulse should look the same except the position the magnitude of the last pulse train should at least have 200 milliamperes to start with. If the gate pulse of any gate is not good, change the PAT card and test again.
6. To place the drive back to normal mode by moving the XTFA board JF connector to receptacle JE.
7. Reset the NFCE card and start the drive.

II. If the drive is shutdown and the fault indicating light is on, it means that any one of the conditions for which the Fault Signal illuminates causes the fault. Check the LED for "telltale" illumination and correct the error before resetting the NFCE card, and subsequently the drive.

III. When the IMOK light (yellow LED) is off in any electronic cards, it indicates that the card is either in a fault condition or that a monitored op-amp output has reached a fault saturation level. The IMOK LED does not latch in the tripped (off) state.

CHAPTER 8

ADJUSTMENTS

8-1. THIS BASIC PROCEDURES FOR MAKING ADJUSTMENTS

This chapter gives basic proce dure for making adjustments to various circuits found in the elementary. No adjustments should be necessary after the drive has been operating properly, even if a care is replaced as the variable components are on a moveable (daughter) board that can be moved to the new card.
Most of the following adjustments are already included in the electronic module elementary 237B6034, but are duplicated here for further clarity.

8-2 Time Delay Circuits
8-3 Speed Feedback and Speed Meter
8-4 Comparators
8-5 Rotor Volts Feedback
8-6 Translating Isolator
8-7 Remote Reference
8-8 Local Reference
8-9 Timed Rate Reference
8- 10 Current Feedback
8-l 1 Thermal Model
8-12 Voltage Feedback (VFB)
8-13 Current Regulator
8-14 Speed Regulator

Turn on control power and turn the bridge power off.

8-2. TIME DELAY (SH.CC)

a. Measure TP6-1K, if it is at logic "0", jumper pin 6-1K to ACOM, external trigger at TP6-1K, remove jumper, TP1-K should change logic level after one second.
If TP6-1K is at logic "1", then external trigger at TP6-1K jumper pin 4-1K to ACOM. TP1-1K should change logic level after one second.
Adjust R4-1K for the right time delay, Remove jumper.

b. Time Delay For Fault Logic (sh. BG)

1. Measure TP6-1 J, if it is at logic "0", jumper pin 6-1J to ACOM, external trigger at TP6-1J, remove jumper, TP1-1J should change logic level after 0.5 seconds.

   If TP6- 1 J is at logic "1", then external trigger at TP6-1J jumper pin 4-1J to ACOM. TP1-1J should change logic level after 0.5 seconds.
   Adjust R1-1K for the right time delay. Remove jumper.
2. Measure TP6-1G, if it is at logic "O", jumper pin 6-1G to ACOM, external trigger at TP6-1G, remove jumper, TP1-1G should change logic level after 3 seconds.

   If TP6-1G is at logic "l", then external trigger at TP6-1G jumper pin 4-1G to ACOM. TP1-1G should change logic level after 3 seconds.

   Adjust R4-1G for the right time delay. Remove jumper.
3. Measure TP6-1H, if it is at logic "O", jumper pin 6-1H to ACOM, external trigger at TP6-1H, remove jumper, TP1-1H should change logic level after 1.5 seconds.

   If TP6-1H is at logic "1", then external trigger at TP6-1H jumper pin 4-1H to ACOM. TP1-1H should change logic level after 1.5 seconds.

   Adjust R1-1H for the right time delay. Remove jumper.

8-3. TACH FEEDBACK & SPEED METER ADJUSTMENT (SH. BS, BR)

(Remove the pulse-tach Feedback at TBCL BS12D and BS15D. Note scaling of the tach is 2kHz. = 1000 rpm.

a. Adjust R34-SPRF1 for 0 volts at TP-15-1G.
b. Input 2kHz. at TBCL BS15D (-) and BS12D (+), adjust R31-SPRF1(blue pot on top of SPRF) (SPD FDBK) so that TP15-1G reads + 10.0 volts. Adjust R6-1J so that TP12-1D reads +10.0 volts and speed meter on door reads 1000 rpm. Adjust the input REF to 1kHz. TP15-1G should read +5.0 volts and TP12-1D reads +5.0 volts and speed meter reads 500 rpm. Reconnect the tach feedback at TBCL.

8-4. COMPARATORS (SH. BQ)

(Input REF to TP11-1G to Simulate Speed Feedback)

a. Minimum Speed Comparator. Adjust the REF to -10.0 volts. Adjust REF to -5.0 volts. Adjust R3-1H so that LED CPA-1H picks up at -5.0 volts.

b. 55% Speed Comparator. Adjust the REF to -5.5 volts.
Adjust R3-1G so that LED CPA-1G picks up at -5.5 volts.

c. 94% Speed Comparator. Adjust the reference to -9.4 volts.
Adjust R3-1F so that LED CPA-1F picks up at -10.0 volts.

d. 15% Current Detector. Adjust R2-1G so that TP4-1G reads -0.75 volts

8-5. ROTOR VOLTS FEEDBACK (SH. BT)

a. Remove R18-1F. Input signal +7.17 Volts at pin 7(-) and 8(+) at JBB. Tie pin 3 and 4 of JBB together. TP10-1E should read the same.
Jumper pin 31-1J to DCOM. Adjust R6-1F so that TP12-1F reads 0.0 volts.
Turn input signal to 0.0 volts. Adjust R4-1F so that TP12-1F reads -10.0 Volts.

b. Replace R18-1F, remove jumpers and signal.

c. When SKD is running at 80% speed, turn RI-IF CCW till TP12-1F reads -8 volts.

8-6. TRANSLATING ISOLATOR (SH. BU)

a. Adjust the speed feedback ref. to -10.0 volts (synchronous speed) pin 4-1L should read + 10.0 volts.

b. Remove jumper at TBCL BU34P and BU36P (if any)

Connect an Ammeter between BU34P and BU36P at TBCL. Ammeter should read 20.0 mA.

Reduce REF to 0.0 Volts. Ammeter should read 4.0 mA. Remove ammeter.

c. At TBCL connect BU08D to BU34P and BU14D to BU36P. Set REF TO -5.0 Volts. TP8-1L should read 5.0 volts.

Set REF to -10.0 volts. TP8-1L should read 10.0 volts.

8-7. REMOTE REFERENCE (SH. BV)

a. Minimum Speed Clamp. Connect a test reference to TP11-1E. Set REF until TP12-1E reads 0.0 volts. Adjust R4-1J until TP12-1J reads 5.0 volts.

b. Maximum Speed Clamp. Adjust REF 1 until TP12-1E reads 10.0 volts. Adjust R6-1H until TP12-1J reads 9.85 volts.

Adjust REF til TP12-1E reads 12.0 volts, adjust R1-1G until TP12-1J reads 9.85 volts.

8-8. LOCAL SPEED REFERENCE (SH. BW)

a. With local speed pot full cw, adjust R5-1H so that TP12-1G is minimum. Adjust R6-1G so that TP12-1G reads 4.85 volts.
Adjust R5-1H so that TP12-1G reads 9.85 volts.

8-9. TIMED RATE REFERENCE (SH. BX)

Make sure pin 3 1- 1G is at logic "0". Connect channel 1 of the chart recorder to TP15-1F, input 5 volt step at TP7-1E.

Adjust RH2-1E so that in 20 seconds TP15-1F increases 5.0 volts.

With pin 31-1G at logic "1", input 10.0 volts at TP11-1G (sh. BT36). Adjust R5-1F so that TP15-1F reads 10.0 volts.

S-10. CURRENT FEEDBACK

a. Input Scaling (elem. sh. CB).

1. Remove DTFA from XTFA.
2. Turn SUP-1A fully ccw and R4-1H fully cw.
3. Increase the test ref to 9.0 and turn R4-1H ccw till LED ACOC-2C (NFOD) just lites. Reset NFCE and jumper pin 62-2C to DCOM.
4. Increase the test ref to 10.5 and turn SUP-1A CW till LED SUP-IA just lites.
5. Remove the ref and reset the NFCE card.
6. Remove the jumper at pin 52-2C.

b. Ammeter

1. With 0.0 on CFB-1A, adjust the null on AM so that it reads 0 amp.
2. Apply a (+) ref volts to IA-TP8 of 1A till TP1 (CFB) = -5.OV.
3. Adjust RH3-1D so that TP15-1K reads 5.76 volts and the ammeter reads 1440 amps.

4. Adjust the REF so that TP15-1K reads 10.0 volts and ammeter reads full scale.

S-11. HEAT SINK THERMAL MODEL CALIBRATION (SH. CE)

a. Jumper pin 64-1C to DCOM to ***release*** firing circuit suicide. Set RH1-1D to obtain 6.0 volts at TP-TJ of NFOC.

b. Connect a test REF to TP3-1A (NFCE). Adjust the reference so that TP-CFB on NFCE reads -7.5 volts. This value will be used for the calibration reference for CFB (FWD) in setting up the thermal model in the forward direction. Adjust R1 on NFOC for 8.57 volts at TP-W/C. Turning T1 CW increases output voltage.

c. Jumper pin 13-2B to ACOM. adjust R4 on NFOC for a voltage -3.10 volts at TP-TJ. This voltage should be set at its steady state value. Turning R4 CW increases output in positive direction.

d. Remove jumper from pin 13-1B NFOC, adjust R5 for a voltage -8.19 volts at TP-TJ. This voltage should be set at its steady state value. Turning R5 CCW increases output in positive direction.

e. Jumper pin 13-2B to ACOM. Without disturbing the test REF setting, open the reference switch and allow the voltage at TP-TJ to settle to 0.0 volts. Connect a chart recorder to TP-TJ. Apply a step reference with the reference switch. On NFOC adjust R2 for a voltage A 2.62 volts in one second as shown in Fig. 8-1.

***** NOTE *****

ALLOW VOLTAGE AT TP-TJ TO SETTLE TO 0.0 VOLTS BEFORE EACH STEP. Turning R2 CCW speeds up the response.

Remove jumper from pin 13-2B to ACOM. Leave chart recorder connected to TP-TJ. Without disturbing REF setting, open the reference switch and allow the voltage at TP-TJ to settle to + 6.0 volts. To speed up the settling process, which can be quite slow, momentarily input volts into pin 16-2B toforce TP-1J to + 6.0 volts. Apply reference with the reference switch. On NFOC adjust R3 for a voltage 6.82 volts in 60 **seconds as shown in Fig. 8-2.**

***** NOTE *****

ALLOW VOLTAGE AT TP-TJ TO **SETTLE TO 6.0 VOLTS BEFORE EACH STEP. Turning R3 ccw speeds up the response.**

f. Jumper-pin 23-1F to 24-1F (short out R20-1F), adjust test REF for 12.5 volts at CFB. LED TOL and CTMP should light when

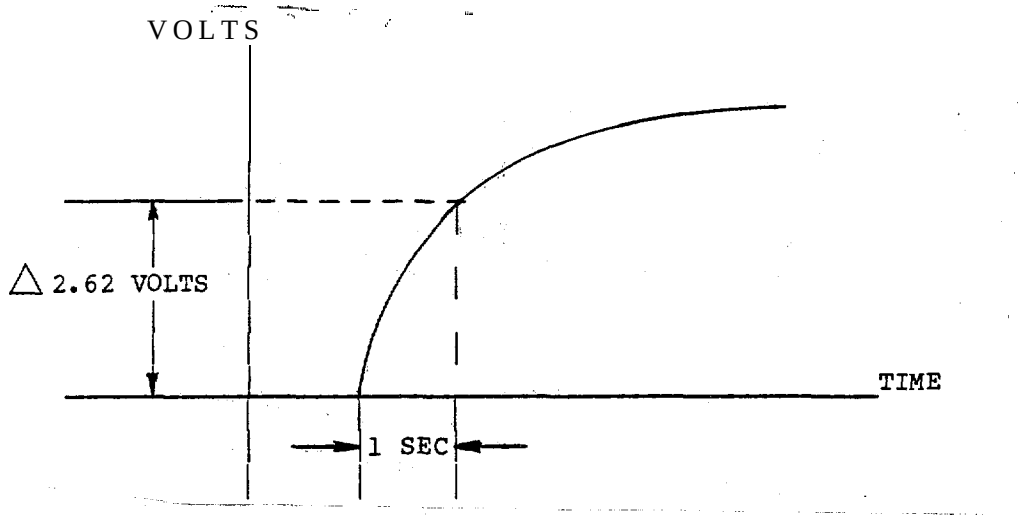


Figure 8-1. Heat Sink Thermal Model - One Second Response

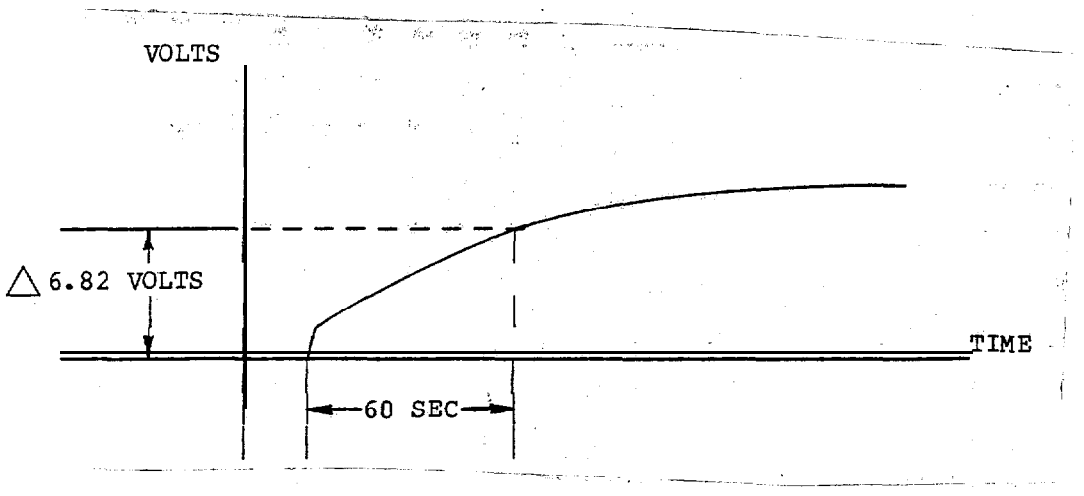


Figure 8-2. Heat *Sink Thermal Model - 60 Second Response*

TP-TJ reaches -7.6 volts. Turn off test REF. LED TOL should latch in and LED CTMP should go out. Reset NFCC card should turn off LED TOL.

S-12. VOLTAGE FEEDBACK (VFB).

This part should be done when drive is running.

a. Input Scaling (elem. sh. ICC)

1. Attach a Simpson between the P and N buses set to read at least 1500 Vdc and adjust null on voltmeter (VM) so that it reads OV.
2. Start motor locally at 500 rpm. Simpson should read 648 Vdc.
3. Adjust VFB-1A so that TP13-1A (NFCE-VFB) = 4.0V. Turning VFB cw increases the output.

b. Voltmeter (VM) (elem. sh. BR)

1. Adjust null when VFB-1A = O.OV so that Vm reads OV.
2. Adjust RH6-1E so that Vm reads 648 volts when both VFB (NFCE cards) = -4.0V. Turning RH4-1C CW increases Vm reading.

8-13. CURRENT REGULATOR (ELEM. SH. BZ)

a. Preliminary Set-Up

1. Adjust R5-1G for + 1.0 volt on pin 29-1G.
2. Jumper pin 6-1E to ACOM, pin 8-1E to ACOM, pin 3 1-1F to ACOM. Jumper pin 65-1F to 66-1F. Adjust R6-1K so that TP4-1E reads -4 volts. Remove all jumpers.

b. Operating Tuning

1. Run motor in SKD mode and do procedure in section 6-7B. Adjust RH5-1E for proper response.

8-14. SPEED REGULATOR (ELEM. SH. BY)

a. Preliminary Set-Up

1. Adjust R4-1C (NSCA-FWD CL) until TP1 = -7.5 volts.
2. Turn R2-1C fully cw.

b. Operational Tune-Up

See Section 6-8.5.